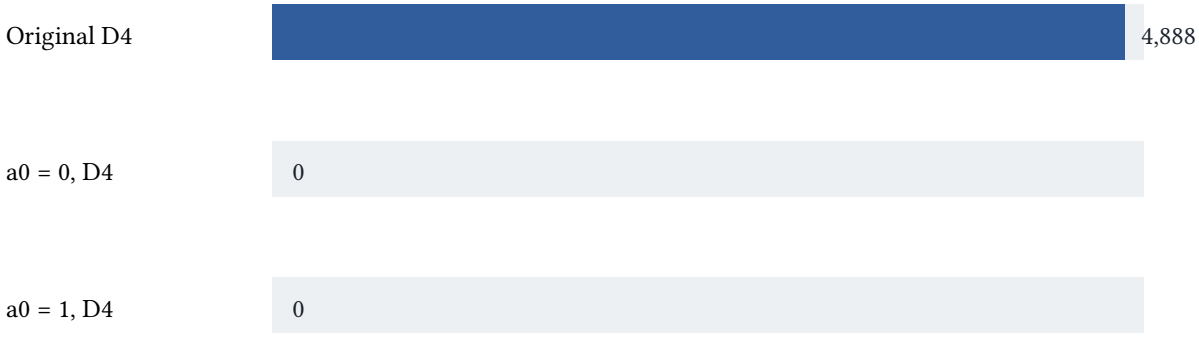
Exact LFD 5 on `fresh-n33-tr0-01`

The named n33 trace-zero Boolean Semaev system has last fall degree exactly **5**. The independent D4 quotient checker verifies that the image of 1 is nonzero and the quotient has dimension at least 4,888. Both restrictions `a0=0` and `a0=1` contain 1 in their degree-four closure. The dossier's branching lemma therefore puts 1 in the original degree-five closure. The exact solution count is zero, so the refutation degree equals the last fall degree for this presentation.

Degree-four quotient dimension

fresh-n33-tr0-01; original is a checked lower bound, branches are exact full-closure results



Unit: F_2 dimension. Both branches: $1 \in W_4$, so codimension 0. Original: $1 \notin W_4$.

Both native branches and both complete independent Python replays found `1 in W4` and the same cumulative dimensions `[1, 34, 562, 6018, 46938]`. The independent code reconstructs the descended system by Möbius interpolation. A separate polynomial-by-polynomial check matched the original and both restricted inputs. Early-stop independent runs found 1 at rank 46,735; their smaller rank reflects their stated stop condition and was preserved in protocol v2. Protocol v3 fixed complete replays before they were run.

The original D4 certificate SHA-256 is `b12d74605a9df9147d89d10191a36fa644cdd5fa70ce610c56e26d870a9fe813`. The raw transcripts and their hashes are indexed by `evidence.json` beside this report. Frozen protocol v3 has a stale `instrument.commit` field for the full independent arms; the actual source is pinned by SHA-256 `d44608877d0731646978a52dd4143d04fe51c0d3359d3317d2bdb9d299b2b875`. The upper bound is independently reproduced exact computation; an explicit standalone membership witness has not been exported.

This named n33 D5 nonmembership attempt should not run: the verified branch computations imply 1 is already in W5. The degree-six search continues on other named inputs, including the still-unclassified n47–51 frontier.